NP-Completeness and Reduction Techniques

How Efficient Transformations Prove Computational Hardness

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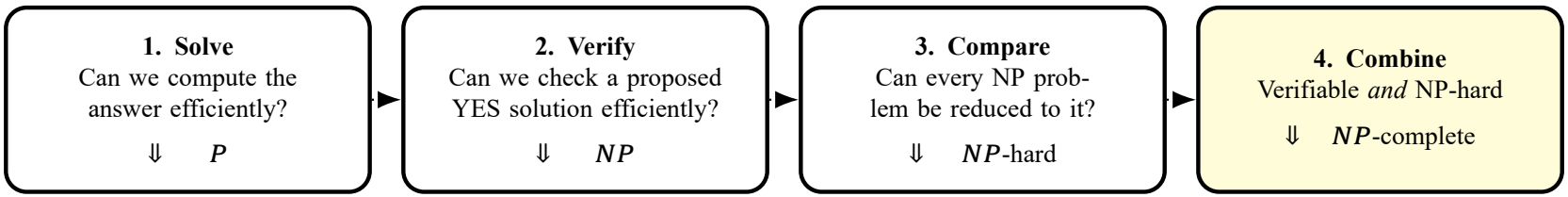
Why Do We Need NP-Completeness?



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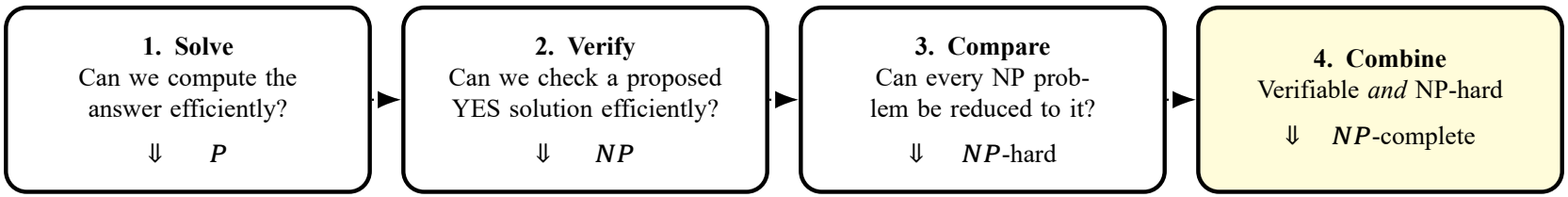

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How can we transfer the known hardness of one problem to a new problem?

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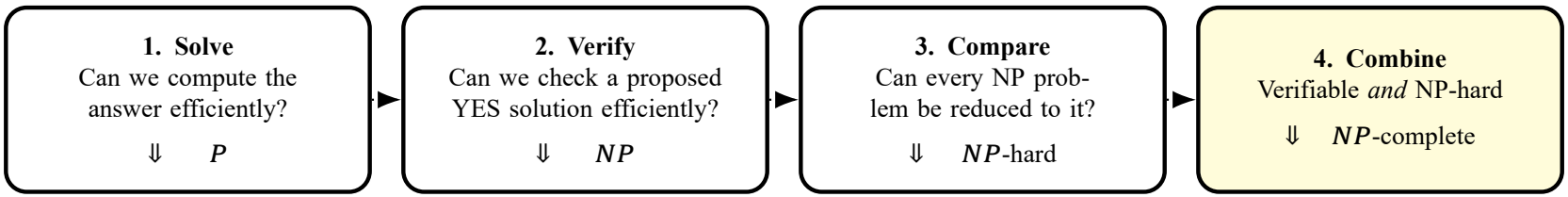

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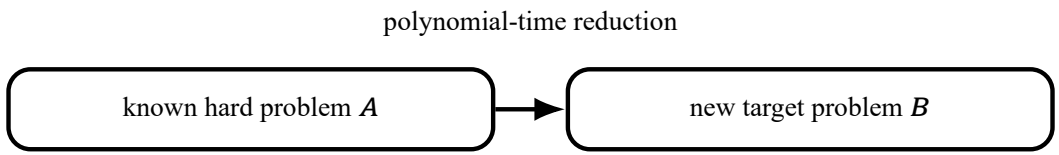
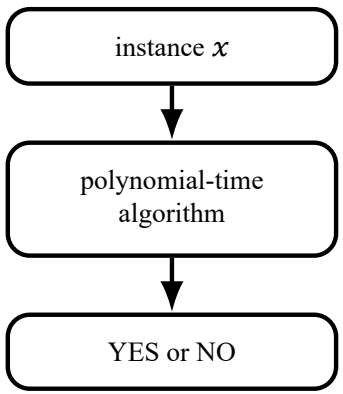


The dotted arrows show the learning sequence, not set containment.

Start with P and NP

P: solve efficiently

Decision problems solvable by a deterministic algorithm in polynomial time.



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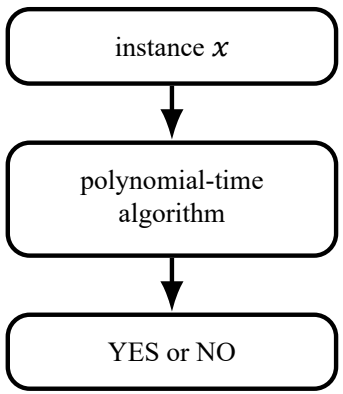

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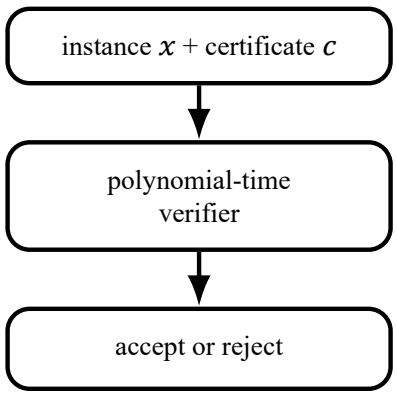
NP: verify efficiently

There are a polynomial-time verifier V and polynomial p such that, for every x,

$$x \in L \iff \exists c : \begin{array}{l} |c| \leq p(|x|), \\ V(x, c) = 1. \end{array}$$



For 3-COLOR, c assigns a color to every vertex; scan all edges to verify it.

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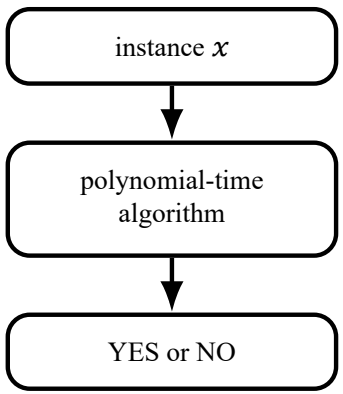

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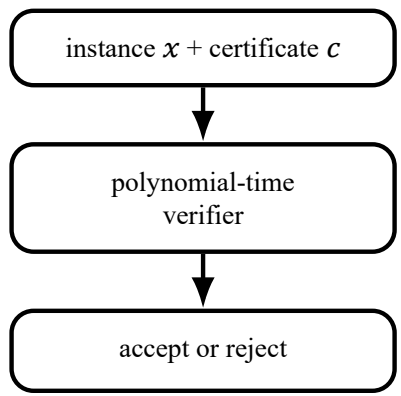

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$P \subseteq NP$: a polynomial-time solver can serve as a verifier and ignore the certificate.

Important: NP means "nondeterministic polynomial time," not "non-polynomial." Whether $P = NP$ is open.

From NP-Hard to NP-Complete

NP-hard: hard for all of NP

A problem H is NP-hard if

$$\forall L \in NP, \qquad L \leq_p H.$$

Here $L \leq_p H$ means an efficient, answer-preserving translation from L to H (formalized next). It need not itself belong to NP.

NP-complete: both properties

$$H \in NP \qquad \text{and} \qquad H \text{ is NP-hard.}$$

These are the hardest efficiently verifiable decision problems.

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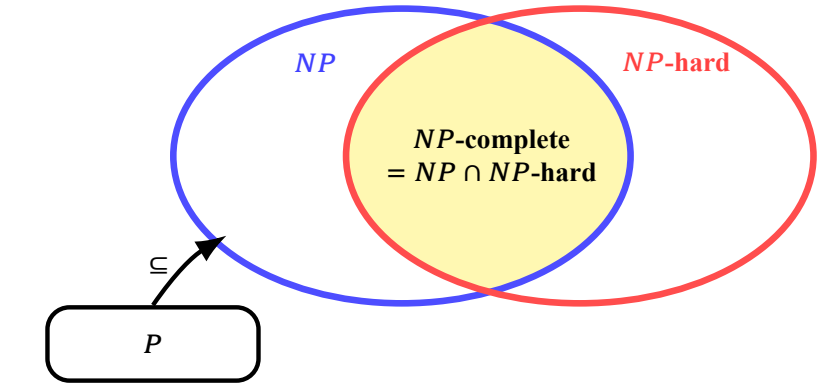

$P \subseteq NP$ is known. Because whether $P = NP$ is open, the exact boundary of P is not drawn.

NP-complete: both properties

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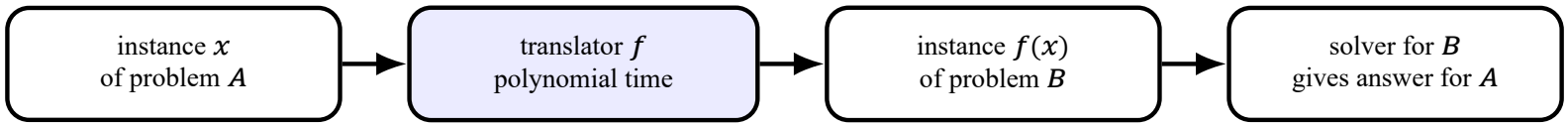
If one NP-complete problem is in P, then $P = NP$.

Polynomial-Time Reduction: an Answer-Preserving Translator



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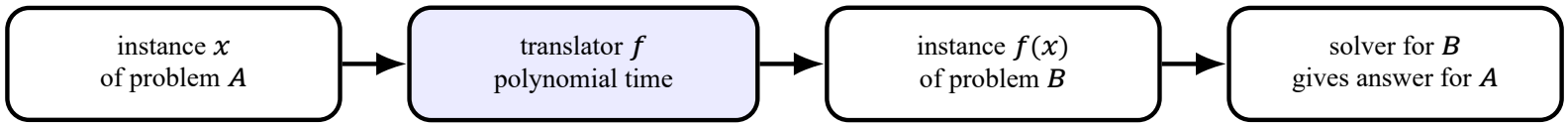

Formal meaning

$A \leq_p B$ means there exists a polynomial-time computable function f such that

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Therefore, a solver for B would also solve A: B is at least as hard as A.

To prove B hard: known hard A $\longrightarrow$ target B.

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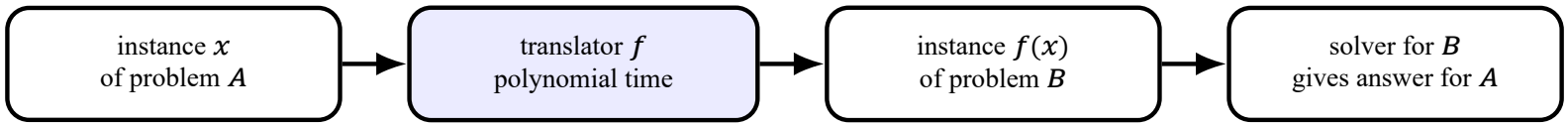

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Tiny example: EVEN $\leq_p$ MULTIPLE-OF-4

Use $f(n) = 2n$. Then

$$n \text{ is even} \Leftrightarrow 4 \mid 2n.$$

n	$f(n)$	preserved?
6: YES	12: YES	YES
7: NO	14: NO	YES

This illustrates the mechanics; a hardness proof starts from a known hard problem.

Reusable Recipe: Proving a Problem NP-Complete

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- State a polynomial-size certificate.
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$$\boxed{\begin{array}{c} A \text{ is NP-hard and } A \leq_p B \\ \Longrightarrow B \text{ is NP-hard.} \end{array}}$$

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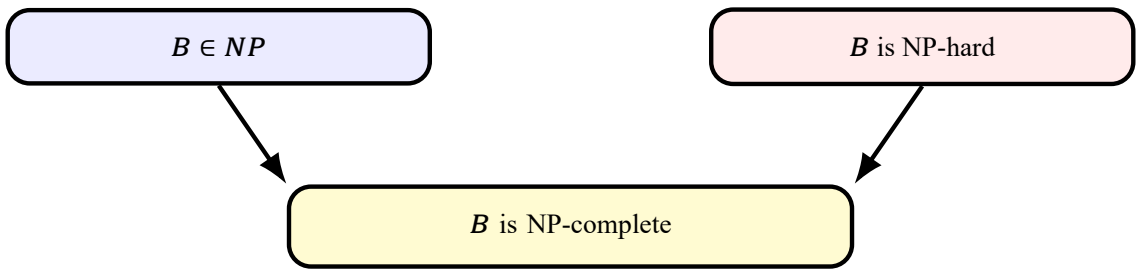

Direction mnemonic: the known hard problem goes *into* the target problem.

Worked Reduction: 3-SAT $\leq_p$ 3-COLOR

Source: 3-SAT

Given a 3-CNF formula (an AND of clauses, each an OR of three literals)

$$\phi = C_1 \wedge \cdots \wedge C_m,$$

is there a truth assignment satisfying every clause? A literal is a variable or its negation.

Target: 3-COLOR

Given a graph G, can its vertices be colored with three colors so that adjacent vertices have different colors?

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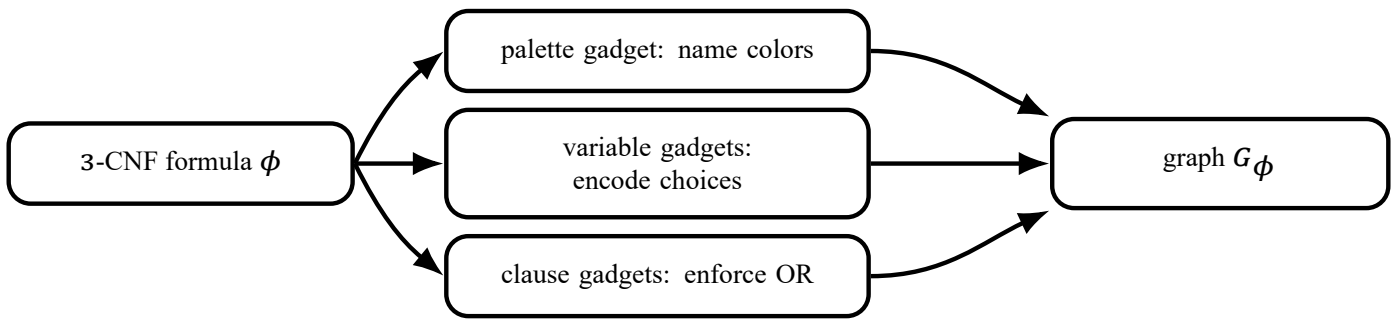

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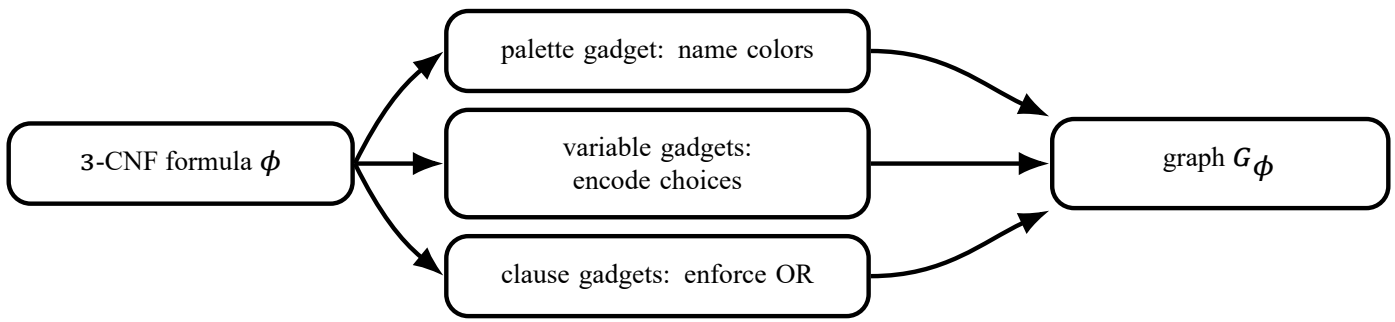


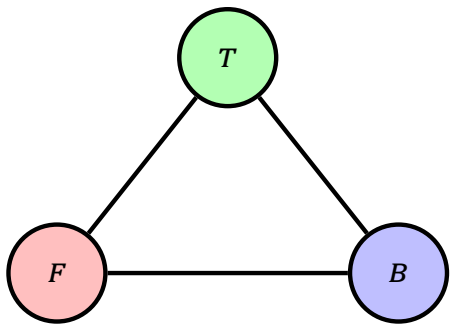
Design invariant: ϕ is satisfiable $\iff$ the constructed graph G_ϕ is 3-colorable.

This is one illustration of the general proof recipe—not the definition of a reduction.

Encoding Truth with Colors

Palette gadget

The triangle forces three distinct colors.
We name them T, F, and B (base/helper).



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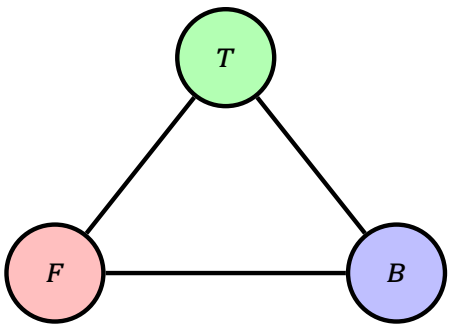

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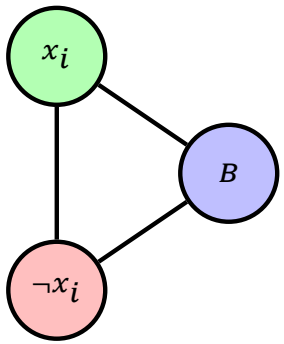
Variable gadget

Connect x_i and $\neg x_i$ to each other and to B.



Exactly two logical colorings:

$(x_i, \neg x_i) = (T, F)$

or

$(x_i, \neg x_i) = (F, T)$.

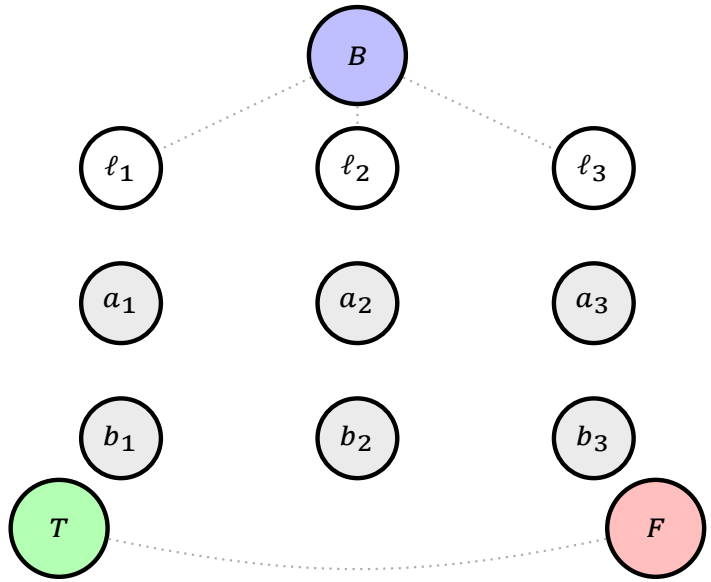
Exactly one of x_i and $\neg x_i$ receives color T.

Constructing the Exact Clause Gadget

For $C = (\ell_1 \vee \ell_2 \vee \ell_3)$, add six fresh vertices a_i, b_i and 13 edges:

$(\ell_i, a_i), (a_i, b_i), (T, a_i)$ for $i = 1, 2, 3$,

$(T, b_1), (b_1, b_2)$, $(b_2, b_3), (b_3, F)$.



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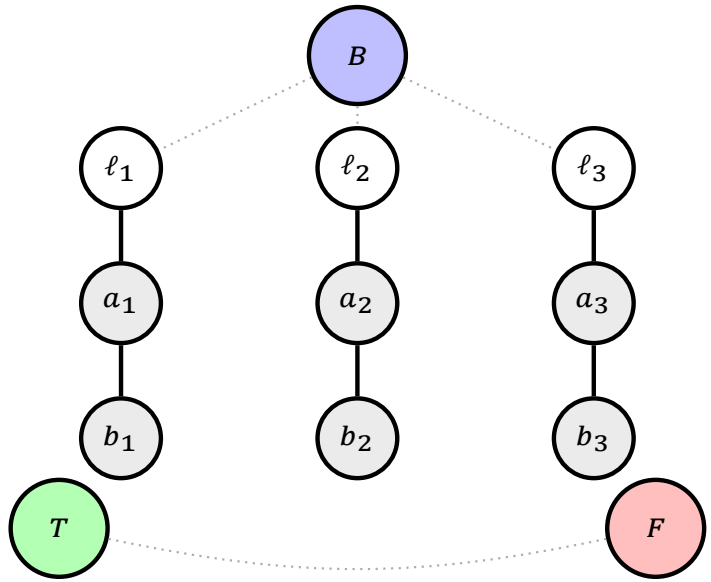

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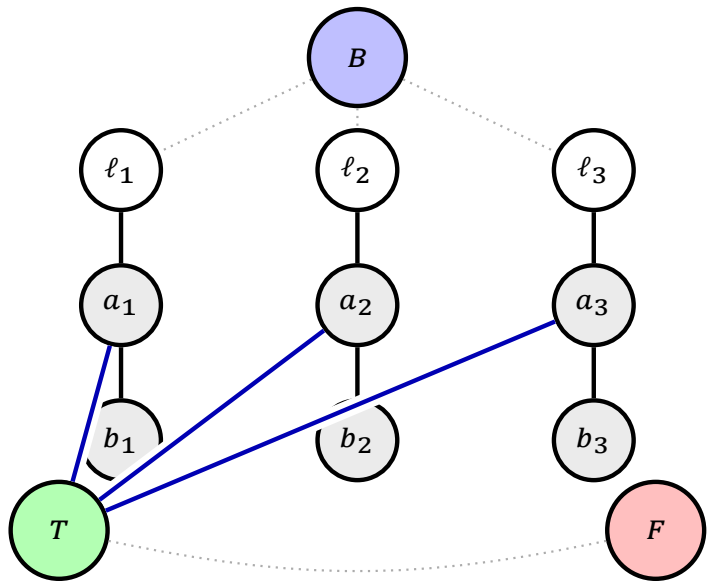

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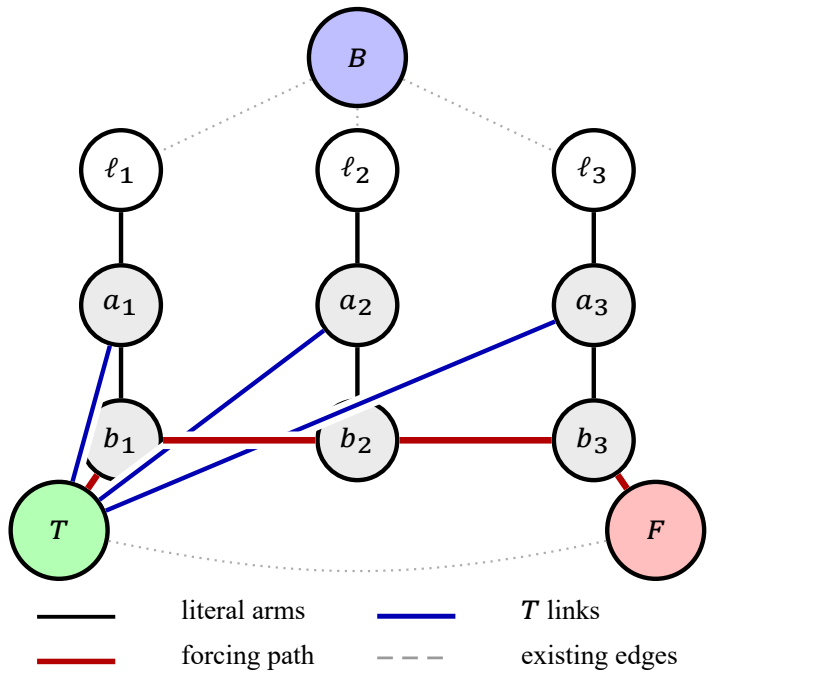
Result: can we complete the coloring?

With inputs fixed in $\{T, F\}$, the internal vertices can be colored exactly when at least one input is T. There is no output vertex.

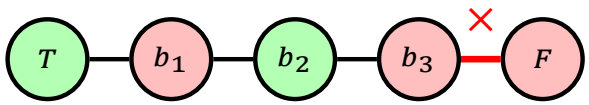


All lines are ordinary edges; line colors only group their roles. Gray/white vertices are unassigned, not extra colors.

Why the Clause Gadget Encodes OR

All inputs are F: impossible

If every $\ell_i = F$, each a_i is forced to B. Hence each b_i can use only T or F.



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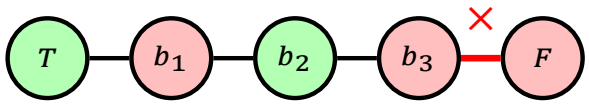

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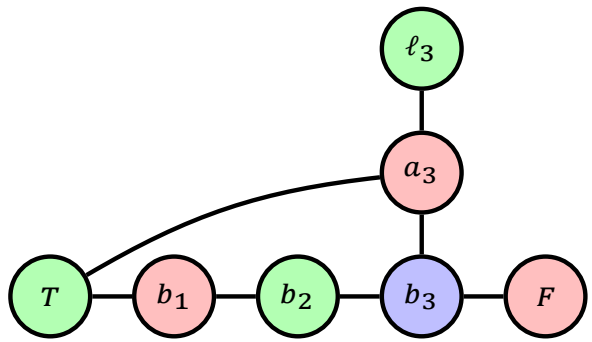

Shown: $i = 3$. For chosen true position $i = 1, 2, 3$, use $(b_1, b_2, b_3) = (B, F, T), (F, B, T), (F, T, B)$, respectively.

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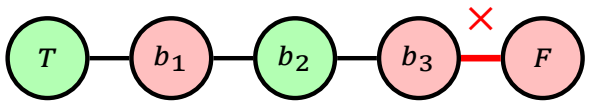

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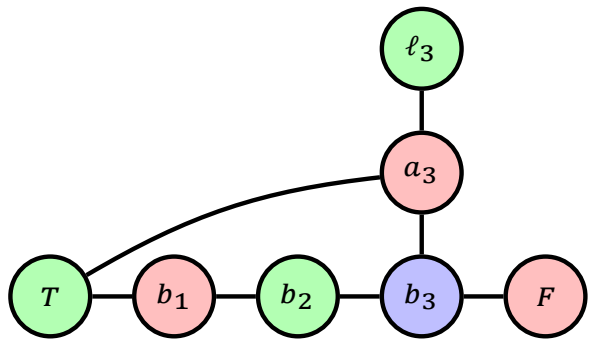

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F, F, F: no extension; all other seven input patterns: an extension exists.

Evaluating One Complete Formula

Use

$$\phi = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee \neg x_3).$$

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Clause input ports

$$C_1 : (x_1, \neg x_2, x_3),$$

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Each clause receives its own six fresh internal vertices.

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One satisfying assignment

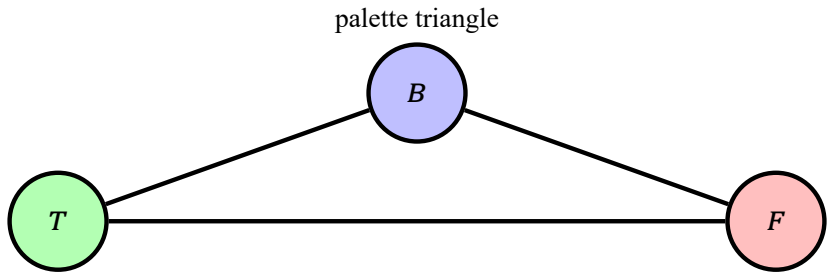
$$x_1 = T, \qquad x_2 = T, \qquad x_3 = F.$$

Thus

$$C_1 = (T, F, F), \qquad C_2 = (F, T, T).$$

Both clause gadgets can be extended.

Assembling the Formula Graph



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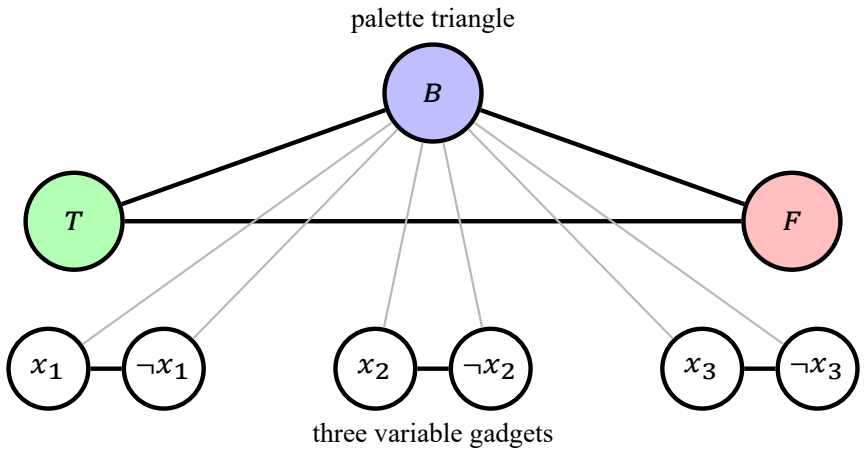

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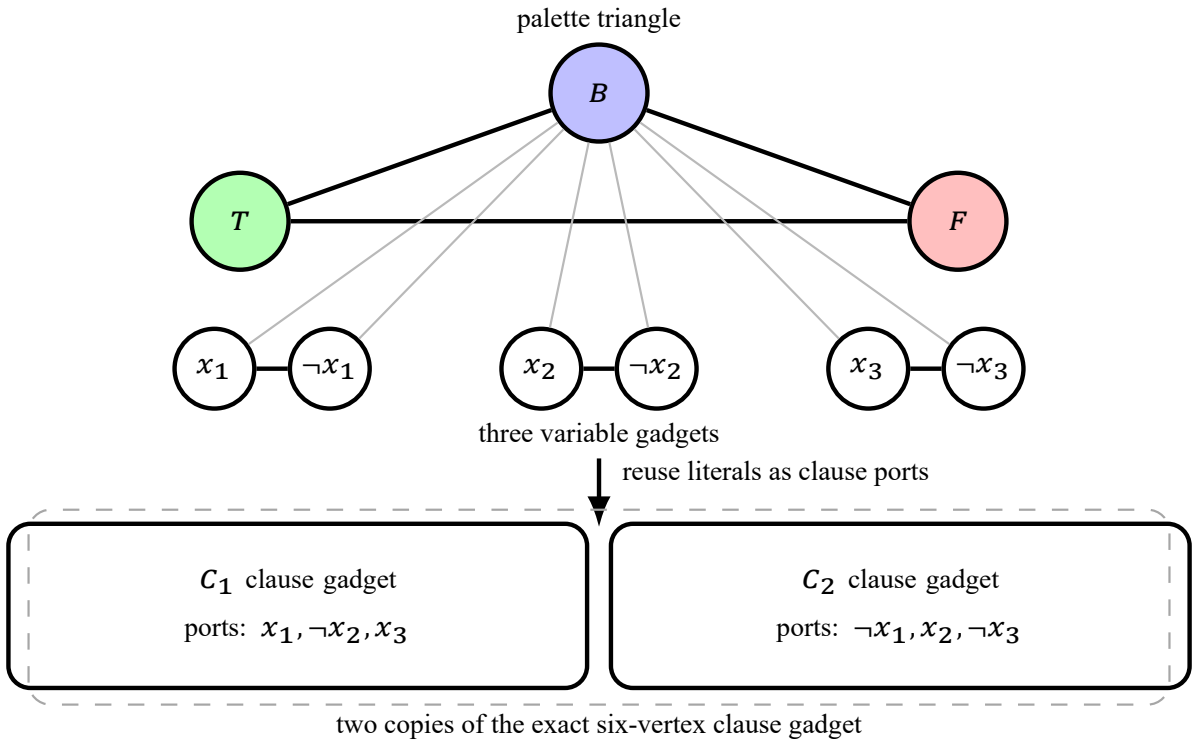

The palette and literal vertices are shared; each clause has fresh internal vertices.

Correctness and Complexity of the Reduction

ϕ satisfiable $\Rightarrow$ G_ϕ 3-colorable

- Give each literal its truth color.
- Every clause has a true literal.
- Therefore every clause gadget extends to a legal coloring.

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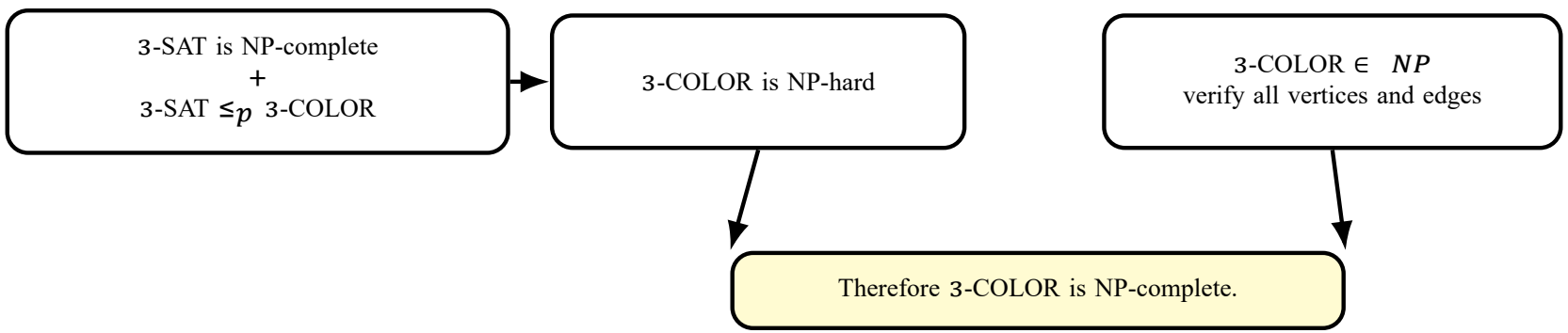

For n variables and m clauses: $|V| = 3 + 2n + 6m$, $|E| = 3 + 3n + 13m$.
Enumerating these vertices and edges takes polynomial time.

A Second Reduction Pattern: 3-SAT $\leq_p$ CLIQUE

$$\phi' = (x_1 \vee \neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_2 \vee x_3) \wedge (x_1 \vee x_2 \vee \neg x_3).$$

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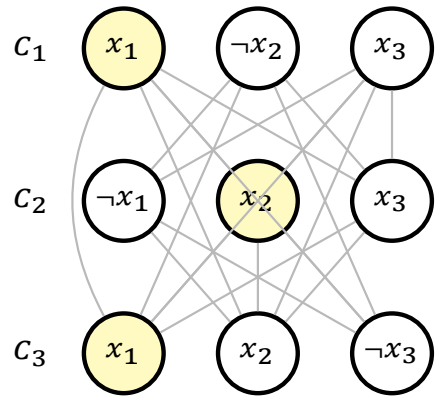

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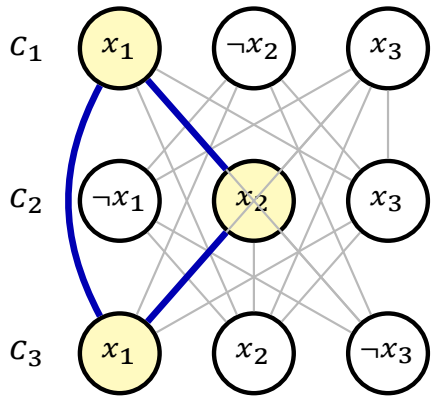

One pairwise noncontradictory literal occurrence from each clause forms a clique.

Conversely, an m-clique must choose one occurrence per clause, with no complementary pair: set them true. Construction: $3m$ vertices and $O(m^2)$ edges.

Takeaway

Classify
P: solve efficiently
NP: verify YES efficiently

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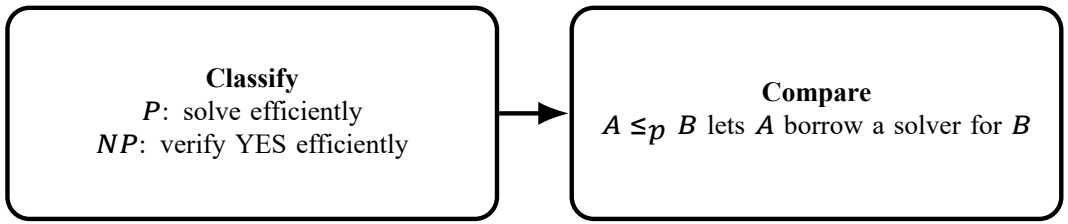
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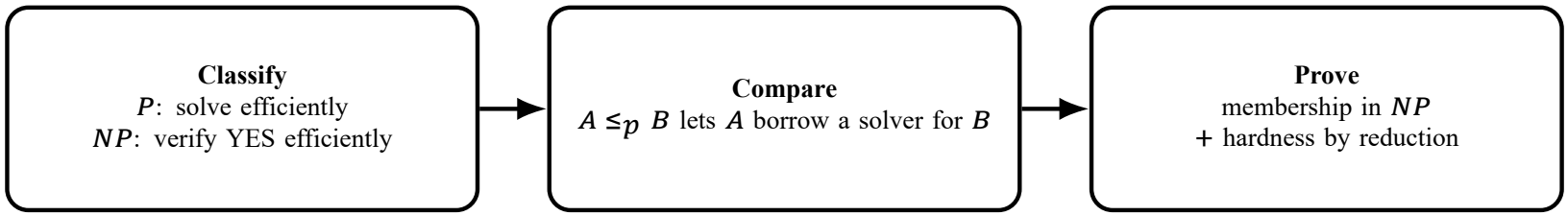
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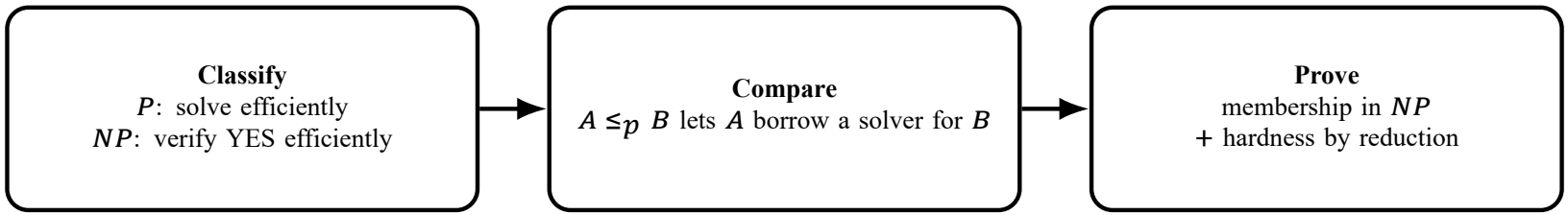
Prove
membership in NP
+ hardness by reduction

Takeaway



One reusable sentence

To prove a target B NP-complete, show $B \in NP$ and reduce a known NP-complete problem A to B in polynomial time while preserving YES/NO answers.

Questions?

Core references: Sipser; Cormen–Leiserson–Rivest–Stein; Princeton COS 423 reduction notes.

References

Primary references

- Michael Sipser, *Introduction to the Theory of Computation*.
- Cormen, Leiserson, Rivest, Stein, *Introduction to Algorithms*.
- CMU 15-451/651 lecture notes on NP-completeness.
- Princeton COS 423 lecture notes, *Polynomial-Time Reductions*.

The exact six-vertex clause gadget and its 13-edge construction follow the Princeton reduction notes.